

Jasper D. Sergio

jdeguzmansergio@gmail.com | Taguig City, Metro Manila | + 639677455508 | <https://github.com/Hekigo11>

Summary

Computer Engineering graduate with hands-on experience in software development, QA testing, systems analysis, computer vision, and embedded systems. Proficient in Python, JavaScript, TypeScript, C/C++, SQL, React, and REST APIs, with experience developing applications, testing software systems, and integrating hardware and software solutions. Adaptable and self-directed, with strong problem-solving, critical thinking, and collaborative development skills.

Experience

Intern - Department of Science and Technology (Information Technology Division) | July–September 2025

Bicutan, Taguig City

QA Tester | July – August 2025

- Executed functional and system testing on internal software applications, validating inputs, workflows, forms, and system functionality against technical specifications.
- Developed and executed test cases and scenarios for multiple system modules, documenting results and identifying functional issues.
- Performed validation checks for data entry, error handling, submissions, and update functionalities to ensure system reliability.

Systems Analyst | August–September 2025

- Studied software requirements and system specifications to understand workflows, functional scope, and user requirements.
- Designed wireframes and interactive prototypes in Figma for user management, task tracking, reporting, and administrative system functions.
- Developed interactive prototype flows including modal interactions, search functionality, dynamic results, and dashboard layouts to communicate system behavior and improve usability.

Freelance Graphic Designer | 2018-2021

Taguig City

- Created graphic designs and visual markups using Adobe Photoshop and Illustrator.
- Incorporated client feedback and revisions to meet design requirements.
- Managed freelance projects from concept to final delivery.

Skills

Languages: Python, JavaScript, TypeScript, C, C++, C#, Java, PHP, SQL, HTML, CSS, Verilog

Frameworks & Libraries: React, Bootstrap, Tailwind CSS, Node JS, Next JS, Vue, ROS2, OpenCV, Pytorch, Tensorflow

Tools & Platforms: Git, GitHub, Vite, MySQL, SQLite, Arduino, PlatformIO, Raspberry Pi, VS Code, Figma, Vivado, GNS3

Technologies: REST APIs, Computer Vision, Machine Learning, Embedded Systems, IoT, Robotics, Database Management

Soft Skills: Adaptable, Self-Taught, Interpersonal, Critical Thinker, Problem-Solving, Project Management, Collaboration, Time Management, Attention to Detail

Projects

NavView: Computer Vision and LiDAR-Based System for Autonomous River Boats | 2025-2026

- Role: Full-Stack Software Developer and Tester
- Led development of an autonomous river-cleaning vessel integrating Python, ROS 2, YOLO, LiDAR, and Arduino for real-time debris detection and obstacle avoidance.
- Trained an AI Computer Vision model with 75% mAP for small trash detection, Developed LiDAR-based obstacle detection, localization, and autonomous navigation.
- Conducted system testing and evaluation to validate real-time perception and autonomous navigation performance.

Marj Food Services E-Commerce Web Application | 2025

- Role: Full-stack Developer
- Led a team to develop an e-commerce web application for powering online storefronts, product/catalog management, catering and order processing using PHP, MySQL, HTML, Javascript, Bootstrap, CSS.

Volcano Monitoring Embedded IOT System | 2025

- Role: Software Developer and Tester
- Built a mobile dashboard using Expo and React Native to ingest real-time sensor data from Firebase Realtime Database, display alert levels and sensor data, and provide controls for sending commands to the system.

Education

Adamson University - Bachelor of Science in Computer Engineering | August 2022 - May 2026

-DOST Scholar

-Academic Merit Awardee

Certifications

- **Introduction to Cybersecurity Essentials** – IBM
- **Cybersecurity Policy Foundations** - Infosec
- **Software Processes and Agile Practices** – University of Alberta
- **Introduction to the Internet of Things and Embedded Systems** - University of California, Irvine
- **Deep Learning for Object Detection** – Mathworks
- **Introduction to Deep Learning & Neural Networks with Keras** – IBM
- **Cloud Fundamentals** – LearnKartS
- **Trust and Security with Google Cloud** – Google Cloud
- **Generative AI: Introduction and Applications** – IBM
- **Generative AI: Prompt Engineering Basics** - IBM